NC STATE UNIVERSITY
Center for Geospatial Analytics

Enhancing the Land Use and Zoning Strategy of Henderson County, North Carolina Through Geospatial Data Modeling

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HENDERSON COUNTY FOUNDED 1838

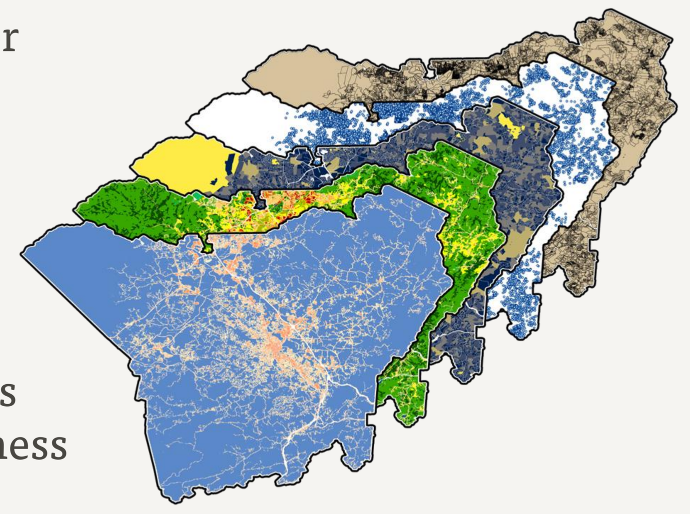
Background

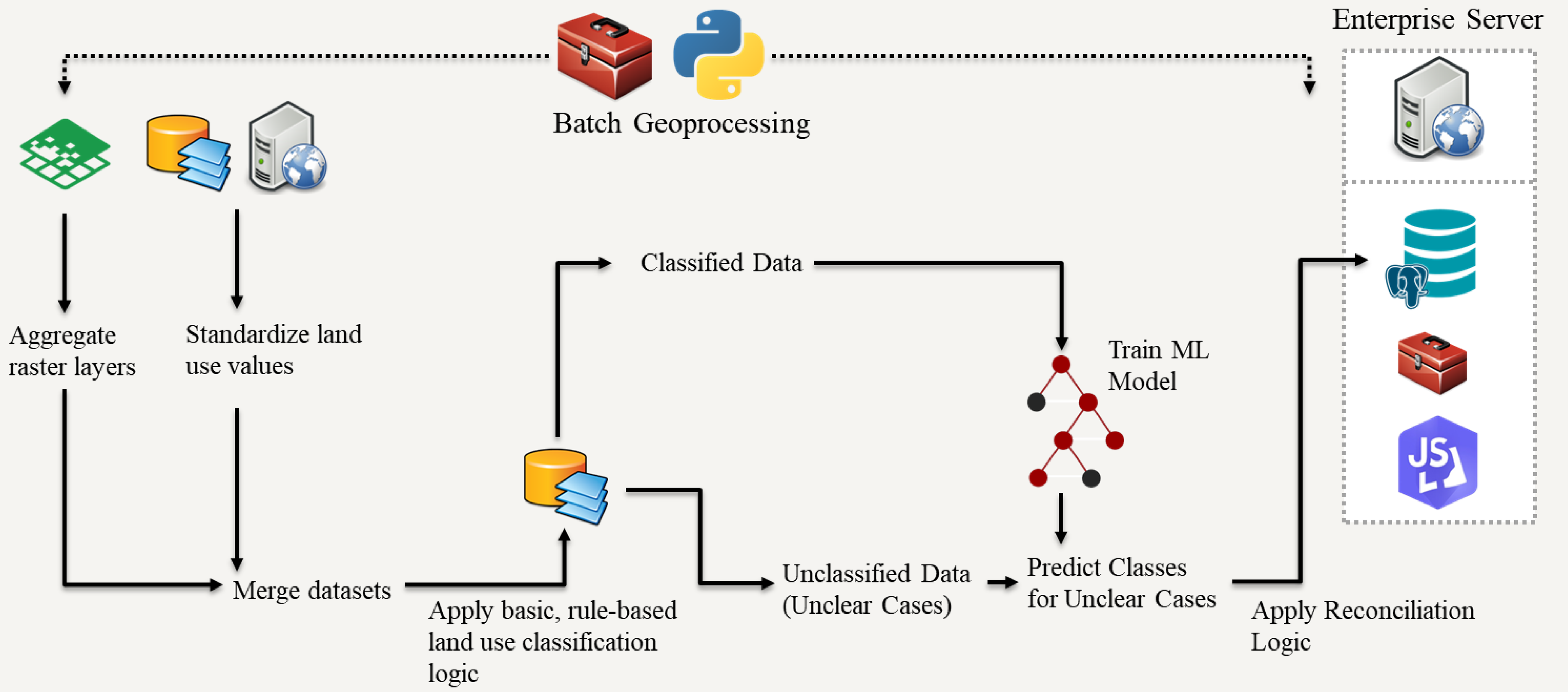
- Henderson County maintains over 70,000 parcel records distributed across multiple internal and external datasets, each using different land use categories; these inconsistencies hinder planning, zoning, and maintenance workflows.
- This project applies rule-based reconciliation and machine-learning classification to establish a single, authoritative land use category for every parcel and delivers the results via an interactive web application to support county planners.

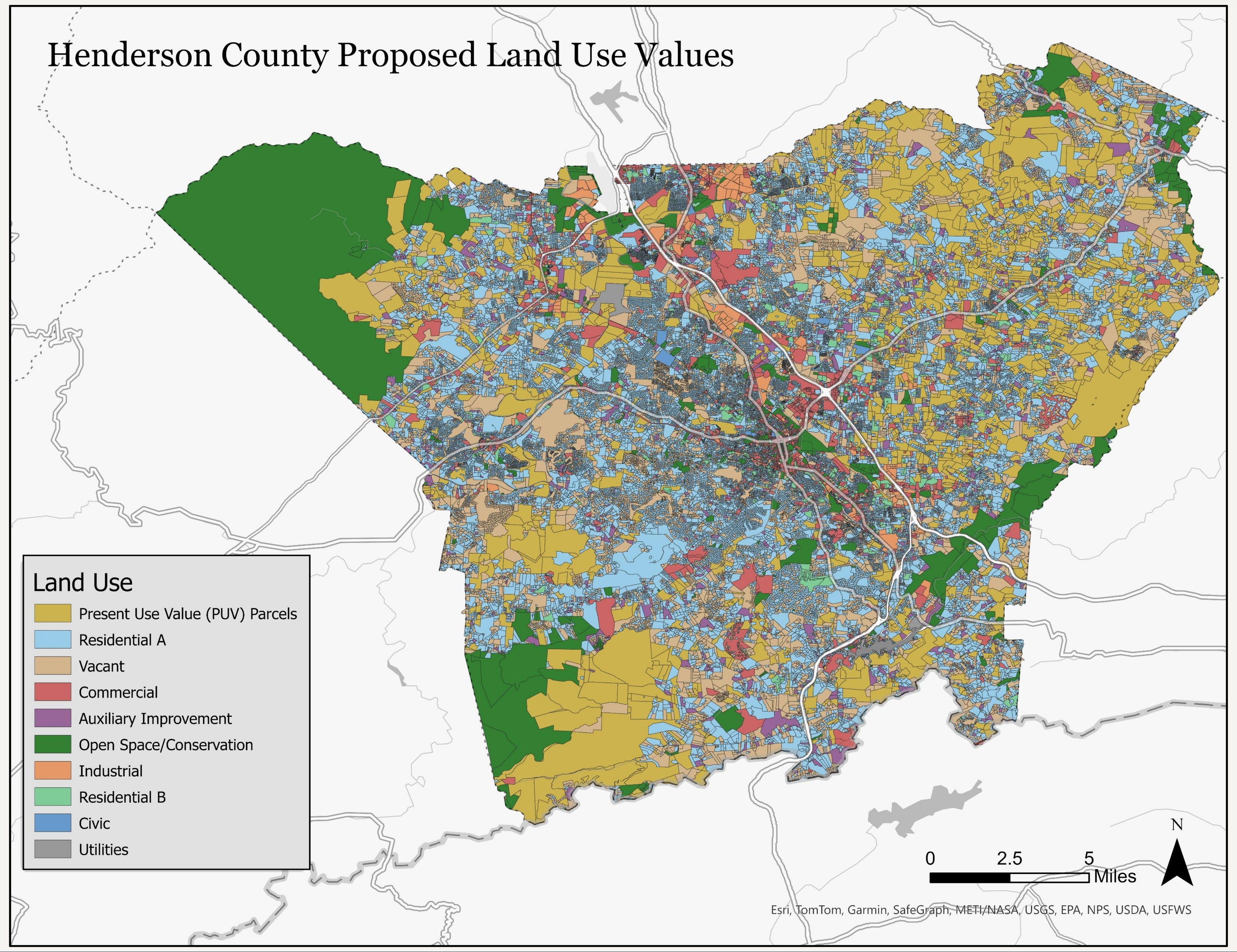
Land Use Categories
Auxiliary Improvement
Civic
Commercial
Industrial
Open Space/Conservation
Present Use Value (PUV) Parcels
Residential A
Residential B
Utilities
Vacant

Methods

1. Data Integration & Rule-Based Classification
   - Aggregated parcel polygons, building footprints, address points, land-cover rasters and external land use polygons into a common coordinate system.
   - Translated all existing land use fields into ten standardized categories.
   - Applied concise business rules (e.g., unanimous source agreement, conservation flags) to classify ~85 % of parcels.
2. Machine-Learning Classification & Reconciliation
   - Trained a supervised model on rule-classified parcels using features such as parcel geometry, building density, land-cover proportions and imperviousness statistics.
   - Predicted labels for the remaining ~15 % of parcels.
   - Resolved conflicts by accepting any model suggestion supported by an original source; otherwise applied a fixed hierarchy (parcel → building → MPO data).
3. Enterprise Geodatabase & Web Application
   - Appended timestamped, authoritative classifications to an enterprise geodatabase.
   - Built an interactive map with the ArcGIS JavaScript API featuring scale-dependent rendering, precomputed SQL views for threshold checks, and a secure geoprocessing service for in-application edits.

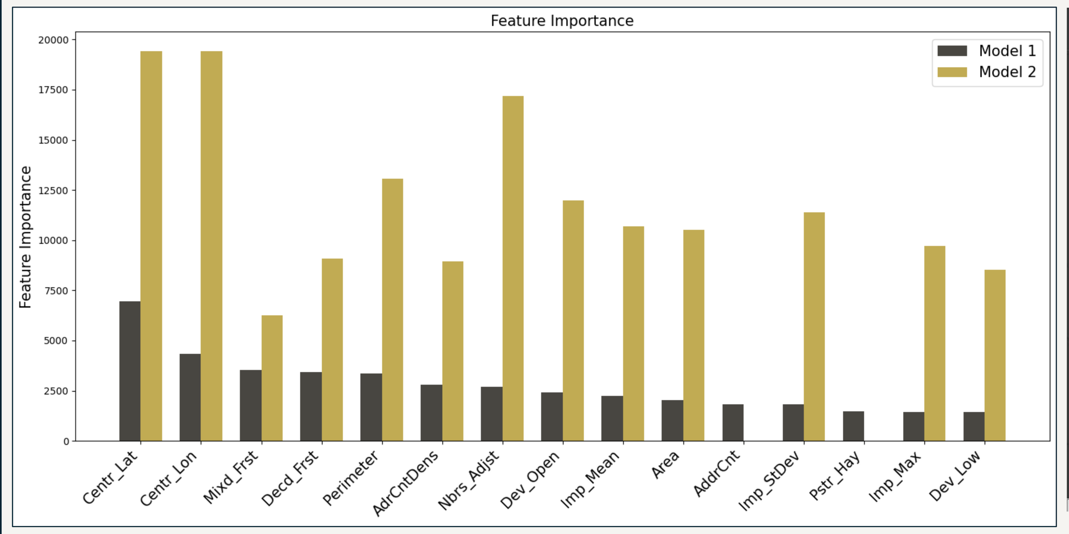


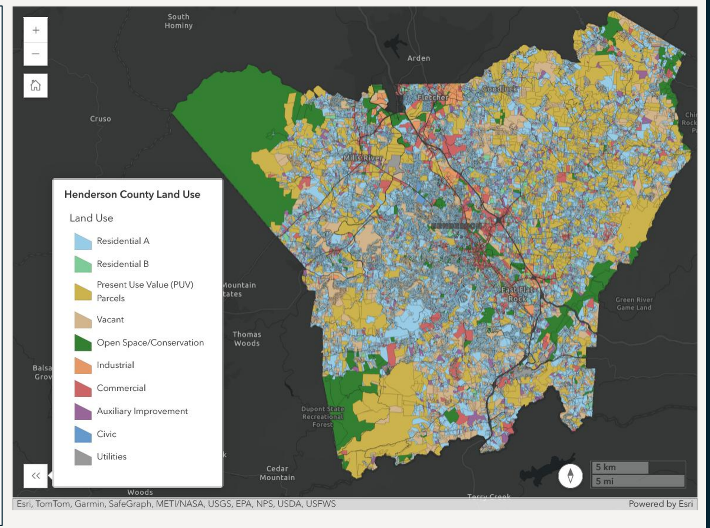




Results

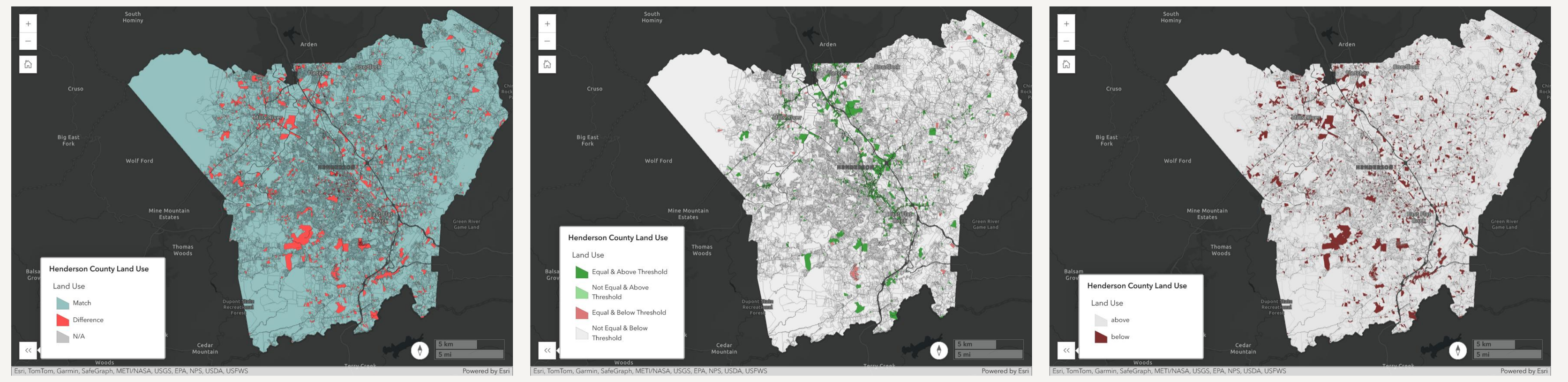
- Classification Coverage: 85 % of parcels labeled by rule-based reconciliation; remaining 15 % resolved by machine learning.
- Model 1 Performance: F1 (weighted) = 0.99, Accuracy = 0.99; leveraged existing parcel, building and MPO label fields to "confirm" ambiguous cases.
- Model 2 Performance: F1 (weighted) = 0.921, Accuracy = 0.927; excluded direct label fields, relying solely on environmental and parcel/building attributes.
- Feature Importance:
  - Although Model 1 can draw on the original label fields, its top fifteen predictors remain spatial and environmental variables (e.g., centroid latitude/longitude, forest cover), since label information is spread across many class-specific fields. In Model 2—where no label fields are included—spatial and environmental features take on even greater relative importance.
  - Model 2 places even greater weight on spatial/environmental features (centroid coordinates, adjacency counts, imperviousness, shape metrics) since it cannot draw on pre-existing label hints.





Discussion

- The hybrid workflow balances transparent business rules with data-driven prediction.
- Rule logic immediately classifies the clear 85 % of parcels, while a two-model ML approach handles the remainder: Model 1 reinforces known labels, Model 2 offers an unbiased cross-check. Discrepancies between the models flag parcels for manual review.
- Precomputing SQL views and using scale-dependent rendering in the web application preserves responsiveness across 70,000+ parcels. Storing only timestamped, finalized classifications in the enterprise geodatabase ensures a concise audit trail while preventing intermediate processing from burdening production services.



Future Research

- Multitemporal Trends: Integrate historical snapshots to monitor land use change trajectories.
- Continuous Model Refinement: Retrain ML models on user-validated edits to improve performance over time.
- Policy Coupling: Link parcel change metrics with socioeconomic and demographic indicators to inform strategic development and conservation planning.

Conclusion

This project delivers a single, authoritative land use layer for Henderson County by uniting disparate source datasets under ten standardized categories.

A reproducible geoprocessing pipeline applies rule-based logic for most parcels and two complementary LightGBM models for the rest—yielding near-perfect agreement with existing records and a robust "second opinion" where sources conflict.

An ArcGIS JavaScript API application then enables planners to explore, validate, and update parcel labels in real time. Together, these components establish a maintainable, data-driven foundation for more consistent zoning, planning and land-management decisions.

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